NURSING HOME STAFFING ANALYTICS

PMP-Aligned Portfolio Project

Project Management | Power BI Analytics | Microsoft Visio

Dataset: CMS Provider Data Catalog (1,000 Facilities, 14 Columns) | Timeline: 1 Mar – 21 Mar 2026

PART 1 Project Charter

Initiation Phase | PMBOK Knowledge Area: Project Integration Management

The Project Charter is the formal document that authorizes a project to exist. It names the project, defines its purpose, identifies responsible roles, and sets high-level expectations for scope, timeline, and deliverables. In PMBOK methodology, no planning activity is formally sanctioned without a signed charter.

1.1 Charter Summary

Project Title	Nursing Home Staffing Quality Analytics: A PMP-Aligned Portfolio Project
Project Sponsor	Portfolio Self-Initiated
Project Manager	Lead Analyst and Project Manager
Project Analyst	Data and Business Process Analyst
Stakeholder Reviewer	Senior Manager, Data and Analytics
Start Date	1 March 2026
Projected End Date	21 March 2026 (21-day sprint)

1.2 Business Case and Problem Statement

Nursing home staffing quality across the United States is a persistent public health concern. The Centers for Medicare and Medicaid Services (CMS) publishes provider-level data on staffing hours, quality ratings, and turnover, yet this data remains underutilized for operational insight at a governance and strategy level. This project applies business process and data analysis techniques to a 1,000-facility sample to surface actionable insights on understaffing patterns, ownership-driven disparities, and weekend care gaps. The project simultaneously demonstrates end-to-end project management capability using PMBOK-aligned principles, Power BI dashboard design at portfolio grade, and Microsoft Visio workflow and lifecycle documentation.

1.3 Project Objectives

- Deliver five analytically rigorous Power BI dashboard pages incorporating DAX measures, drill-through pages, advanced visual types, and conditional formatting.
- Produce a full Visio documentation set: Project Lifecycle Swimlane, Work Breakdown Structure tree diagram, Gantt Chart, and Analytics Workflow Flowchart.

- Demonstrate eight PMBOK-aligned project management artefacts covering all five process groups: Initiating, Planning, Executing, Monitoring and Controlling, and Closing.
- Package all deliverables as a printed portfolio suitable for presentation to a panel of senior analytics professionals in a healthcare context.

1.4 Scope Statement

In Scope: Analysis of NH_Staffing_1000_Clean.csv (1,000 rows, 14 columns, zero nulls). Five Power BI business problem pages plus an Executive Summary page. Four Visio diagrams representing project structure and analytical workflow. All eight PMBOK artefact documents listed in this portfolio.

Out of Scope: Data collection, web scraping, or additional data sources beyond the provided CSV. Statistical modelling or machine learning. Power BI Service deployment or live data refresh. Any facility-level individual data disclosure beyond aggregate analysis. Any additional Power BI pages beyond the defined six-page report.

Scope Creep Controls

Creep Scenario	Risk Level	Control Action
Adding a sixth or seventh business problem beyond BP5	High	Scope is frozen at five BPs. Additional ideas are logged in the Backlog for a future project phase.
Adding data cleaning or transformation steps in Power Query	Medium	The dataset is pre-cleaned. Any transformation request is routed through the Change Control Log.
Building a live Power BI Service dashboard with scheduled refresh	High	Out of scope. Desktop .pbix file only is the defined deliverable.
Expanding analysis to the full 14,710-row CMS dataset	Medium	The 1,000-row stratified sample is sufficient for portfolio demonstration. Logged as Phase 2 backlog.
Adding a fifth or sixth Visio diagram beyond the four defined	Low	One additional diagram permitted only after all four primary diagrams pass validation.

1.5 Project Milestones

Milestone	Description	Target Date	PMP Phase
M1: Charter Approved	Project formally initiated. Scope, objectives, and stakeholder roles documented.	3 Mar 2026	Initiating
M2: Planning Complete	WBS, Risk Register, RACI, Communication Plan, and Milestone Schedule finalized.	10 Mar 2026	Planning
M3: Power BI Report Built	All five BP pages plus Executive Summary complete with DAX and formatting.	16 Mar 2026	Executing
M4: Visio Diagrams Complete	All four Visio diagrams built and reviewed against acceptance criteria.	17 Mar 2026	Executing
M5: Portfolio Validated	All outputs pass validation checklist. Issue log closed. Print-ready.	19 Mar 2026	Monitoring
M6: Portfolio Delivered	Three printed copies assembled for panel presentation.	21 Mar 2026	Closing

PART 2 Project Management Plan

Planning Phase | Scope, WBS, Risk, Stakeholders, RACI, Communication

The Project Management Plan is a collection of subsidiary plans that together govern how the project will be executed, monitored, and closed. Each subsidiary plan addresses a specific PMBOK knowledge area. The key planning artefacts for this portfolio project are documented in sections 2.1 through 2.7 below.

2.1 Work Breakdown Structure (WBS)

The WBS decomposes total project scope into progressively smaller, manageable components. Each element produces a tangible deliverable. The tree diagram below presents the full WBS hierarchy visually, with WBS codes embedded in each shape and colour-coding applied by phase. This format is the PMBOK-recommended presentation for a WBS and is significantly more readable than a flat table when presented to stakeholders.

Visio Build Instructions: File, New, Business, Org Chart Wizard or Basic Flowchart. Place the root node at top with the project title and code 0.0. Add five Level 1 child shapes below for each PMP phase. Add Level 2 grandchild shapes under each phase for individual WBS work packages. Include WBS codes inside each shape. Use Auto Layout under Home, Position, Auto Layout to balance the tree. Apply consistent phase colours using Format Painter. The result should visually mirror the reference diagram below.

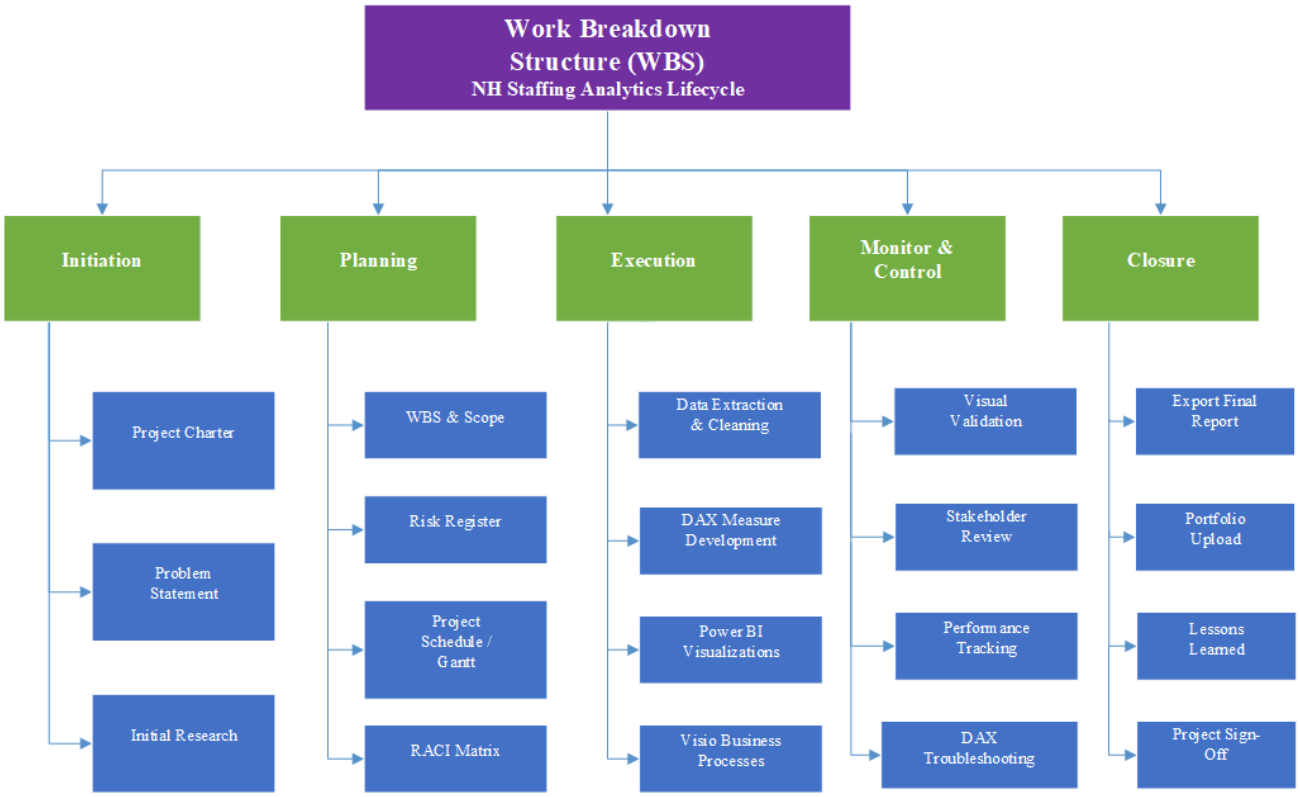

Figure 1: WBS Hierarchical Tree Diagram - Build in Visio using Org Chart or Basic Flowchart template

Deliverable	Duration	Responsible Role
Initiation	3 days	Project Manager
Project Charter	1 day	Project Manager
Stakeholder Register	0.5 days	Project Manager
Scope Statement and Creep Controls	0.5 days	Project Manager
Planning	4 days	Project Manager and Data Analyst
WBS and WBS Dictionary	1 day	Project Manager
Risk Register	0.5 days	Project Manager
RACI Matrix	0.5 days	Project Manager
Communication Plan	0.5 days	Project Manager
Milestone Schedule	0.5 days	Project Manager
Execution: Power BI	7 days	Data Analyst
Load CSV and build Executive Summary page	1 day	Data Analyst
Build BP1 and BP2 with DAX and formatting	2 days	Data Analyst
Build BP3, BP4, BP5 with drill-through	3 days	Data Analyst
Cross-page slicers, bookmarks, and dynamic titles	1 day	Data Analyst
Execution: Visio	3 days	Business Process Analyst
Project Lifecycle Swimlane Diagram	1 day	Business Process Analyst
WBS Tree Diagram	0.5 days	Business Process Analyst
Gantt Chart (21-day sprint)	0.5 days	Business Process Analyst
Analytics Workflow Flowchart	1 day	Business Process Analyst
Monitoring and Control	2 days	Project Manager
Validate all Power BI visuals against acceptance criteria	1 day	Data Analyst and Project Manager
Issue log review and change control review	0.5 days	Project Manager
Final cross-check against scope statement	0.5 days	Project Manager
Closure	2 days	Project Manager
Export Power BI visuals as PNG for print	0.5 days	Data Analyst
Print and assemble portfolio (3 copies)	0.5 days	Project Manager
Lessons Learned document	0.5 days	Project Manager
Verbal walkthrough rehearsal	0.5 days	Data Analyst

2.2 Risk Register

Each risk is scored on Probability and Impact (scale 1–3). Risk Score is the product of the two, giving a Priority ranking of Critical (9), High (6), Medium (4), or Low (1–2).

ID	Risk Description	Prob.	Impact	Score	Response Strategy
R01	Scope creep: additional business problems added beyond BP5	High	High	9	Freeze scope at five BPs. Route all additions to Change Control Log.
R02	DAX measure returns incorrect values due to wrong column reference	Med	High	6	Validate each measure in a Table visual before embedding in chart.
R03	Printer quality issue on interview day	Med	High	6	Print the day before. Maintain a PDF backup accessible on a mobile device.
R04	Visio diagrams take longer than estimated due to layout formatting	Med	Med	4	Use built-in Visio themes. Cap formatting time at 30 minutes per diagram.
R05	Power BI file corrupts or is lost	Low	High	3	Save to OneDrive and USB daily. Maintain a versioned backup copy.
R06	Panel asks about analytical choices outside prepared answers	Med	Med	4	Prepare a two-sentence rationale for each business problem and chart type.
R07	Dataset values appear unexpected after loading into Power BI	Low	Med	2	Pre-verified: 1,000 rows, zero nulls. Anomalies are documented as findings.

2.3 Assumption Log

The Assumption Log is a PMBOK-recognized artefact that documents the assumptions underlying the project plan. Undocumented assumptions are a leading cause of scope disputes and rework because they represent risks that have not been formally acknowledged.

Assumption ID	Assumption Statement	If Assumption Proves False
A01	The CMS 1,000-facility sample is representative of national staffing quality trends	Analysis scope must be qualified as sample-based and not nationally generalizable
A02	All 14 columns in the dataset are correctly defined per the CMS data dictionary	Column definitions must be re-verified against the CMS Provider Data Catalog
A03	Power BI Desktop is available and licensed for use on the project workstation	An alternative BI tool or Power BI trial must be arranged before execution begins

A04	Microsoft Visio is available for building all four required diagrams	A free Visio alternative such as draw.io may be substituted with quality review
A05	A colour printer will be available for portfolio assembly on or before 21 Mar	A commercial print service must be arranged at least two days before the deadline
A06	The interview panel has basic familiarity with PMP terminology and PMBOK concepts	A brief 60-second context statement must be added to the verbal walkthrough introduction

2.4 Stakeholder Register

Role	Area	Interest Level	Influence	Engagement Strategy
Senior Manager, Data and Analytics	Project Review and Evaluation	High	High	Lead with analytics maturity framing and governance language throughout walkthrough
Manager, Data and Analytics	Technical Evaluation	High	High	Prepare to discuss DAX logic, chart design rationale, and visual selection decisions
Project Manager, Data and Analytics	PM Capability Evaluation	High	High	Open the walkthrough with the Gantt and WBS to immediately establish PM credibility
CMS (Data Source Owner)	Data Provenance	Neutral	Low	Cite CMS as the data source on all printed visuals for data governance transparency
Lead Analyst and Project Manager	Project Delivery	High	High	Self-managed. Daily personal check-ins against the WBS and milestone schedule

2.5 RACI Matrix

R = Responsible (performs the work). A = Accountable (owns the outcome). C = Consulted (provides input before decisions). I = Informed (kept updated after decisions).

Deliverable	Project Manager	Data Analyst	Business Process Analyst	Reviewer / Panel
Project Charter	R / A	I	I	I
Scope Statement and Creep Controls	R / A	C	C	I
WBS and WBS Dictionary	R / A	C	C	I
Risk Register	R / A	C		I

Assumption Log	R / A	C		I
RACI Matrix	R / A	I	I	I
Communication Plan	R / A	I	I	I
Change Control Log	R / A	C	C	I
Load Dataset and Verify Data	A	R		C
Power BI Executive Summary	A	R		C
Power BI BP1 Dashboard	A	R		C
Power BI BP2 Dashboard	A	R		C
Power BI BP3 Dashboard	A	R		C
Power BI BP4 Dashboard	A	R		C
Power BI BP5 Dashboard	A	R		C
Visio Lifecycle Swimlane	A	I	R	C
Visio WBS Tree Diagram	A	I	R	C
Visio Gantt Chart	A	I	R	C
Visio Workflow Flowchart	A	I	R	C
Validation Checklist Review	R / A	C	C	I
Lessons Learned Document	R / A	C	C	I
Print Portfolio Assembly	R / A	C	C	

2.6 Communication Plan

The Communication Plan defines what information will be shared, with whom, through what channel, and how frequently. This artefact demonstrates to evaluators that stakeholder communication is treated as a structured activity with defined ownership rather than an informal afterthought.

Communication	Audience	Frequency	Channel	Responsible Role
Project status self-review against WBS	Project Manager (self)	Daily during sprint	Personal task log or OneNote	Project Manager
Milestone completion check	Project Manager (self)	At each milestone date	WBS tick-off in this document	Project Manager
Change log update	Project record	As changes arise	Change Control Log (Section 2.7)	Project Manager
Assumption log review	Project record	At Planning and Monitoring phase	Assumption Log (Section 2.3)	Project Manager
Final portfolio delivery	Interview Panel	Once on interview day	In-person printed portfolio	Lead Analyst
Verbal walkthrough rehearsal	Lead Analyst (self-practice)	Days 20 and 21	Spoken aloud against printed copies	Lead Analyst

2.7 Change Control Log

Every change to baseline scope, schedule, or deliverables must be logged here before implementation. This ensures the project record remains accurate and auditable. Even approved changes that seem minor create a precedent if they are not documented, which can undermine scope control.

Change ID	Date	Description	Requested By	Impact Assessment	Decision
CCR-001	8 Mar	Add drill-through page to BP3 scatter chart to show facility-level detail on click	Data Analyst	Low. Adds approximately 2 hours. Within schedule buffer. Improves portfolio depth.	Approved
CCR-002	TBD	[Log all future change requests here before implementing any modification]			Pending

PART 3 Power BI Analytics Dashboard Guide

Execution Phase | Five Portfolio-Grade Business Problems with DAX and Advanced Visuals

Each of the five business problems below is built to portfolio standard. This means every page includes a clear hypothesis, at least two DAX measures, one advanced visual technique beyond a basic bar or column chart, and a drill-through or supporting analytical layer. The report opens with an Executive Summary page that acts as the navigation hub.

Report Setup: Do This Before Building Any Page

1	**Apply a Custom Theme** View tab, Themes, Customize Current Theme. Set Primary color to #1F4E79, Secondary to #2E75B6. Save as NH_Staffing_Theme.json. Apply to all pages. Consistent colour across the report signals professional intent immediately.
2	**Build the Executive Summary Page** Create a page called Executive Summary. Add four Card visuals in a horizontal row: Total Facilities, Understaffed Count (via DAX), Average Overall CMS Rating, and Average RN Hours. Below the cards add a Slicer for State and a Slicer for Ownership Type. Enable Sync Slicers (View, Sync Slicers) so both slicers filter every page in the report.
3	**Create Bookmarks** View tab, Bookmarks. Create a bookmark called Default View before building visuals. Create a bookmark called BP[n] Complete after each page is finished. This enables report resetting during a live presentation and demonstrates knowledge of interactive Power BI features.
4	**Add Dynamic Page Titles via DAX** On each page add a Text Box whose content is driven by a DAX measure. Example: SELECTEDVALUE([State], "All States") concatenated with " - Staffing Analysis". This makes the report title respond to the slicer selection, signalling a live and responsive report rather than a static output.

BP1 Geographic Understaffing Risk Map

Business Question: Which states show the highest proportion of facilities below the CMS 3.48-hour benchmark and does this correlate with lower overall CMS ratings?

Key Columns: State, Staffing Status, Total Nurse Hours Per Resident Day, Overall Rating

Hypothesis: States with more than 40% of facilities understaffed are expected to also carry lower average CMS ratings, suggesting systemic rather than isolated staffing failures.

DAX Measures

Understaffed %	`Understaffed Pct = DIVIDE(COUNTROWS(FILTER('NH_Staffing_1000_Clean','NH_Staffing_1000_Clean'[Staffing Status]="Understaffed")),COUNTROWS('NH_Staffing_1000_Clean'),0)`

Below Benchmark	`Below Benchmark Flag = IF(AVERAGE('NH_Staffing_1000_Clean'[Total Nurse Hours Per Resident Day])<3.48,"Below CMS Benchmark","At or Above")`

Build Steps

1	**Choropleth Map** Select Map or Shape Map. Drag State to Location. Drag Understaffed Pct to Color Saturation. Add a tooltip field for Average Overall Rating so hovering over a state shows both dimensions simultaneously without requiring a drill-through.
2	**Dual-Axis Bar Chart** Clustered Bar. Y-axis: State (top 15 by Understaffed Pct). Primary X: Understaffed Pct. Secondary axis: Average Overall Rating as a Line series. This tests the hypothesis that high understaffing correlates with low CMS ratings in a single view.
3	**Drill-Through Page: State Detail** New page called State Detail. Table with Facility Name, Staffing Status, Total Nurse Hours, RN Hours, and Overall Rating. Right-click the State field and enable as Drill-Through. Users can click any state bar and drill directly to individual facility data.
4	**KPI Card with Context Label** Card showing Understaffed Pct. Below it add a Text Box: CMS Recommended Minimum: 3.48 Hours per Resident per Day. This contextualizes the percentage for viewers unfamiliar with the CMS standard.

BP2 RN Coverage Disparity by Ownership Type

Business Question: Does For-Profit ownership structurally produce lower RN hours and does this gap persist across all facility size bands?

Key Columns: Ownership Type, RN Hours Per Resident Day, Total Nurse Hours Per Resident Day, Facility Size

Hypothesis: For-Profit facilities will show both lower absolute RN hours and a higher aide-to-RN ratio, suggesting systematic cost-substitution of registered nurses with lower-cost aide staffing.

DAX Measures

RN Share	`RN Share of Total Hours = DIVIDE(AVERAGE([RN Hours Per Resident Day]),AVERAGE([Total Nurse Hours Per Resident Day]),0)`

RN Gap	`RN Hours Gap = AVERAGE([RN Hours Per Resident Day]) - 0.75`

Build Steps

1	**Clustered Bar with Small Multiples** Clustered Bar. X-axis: Average RN Hours. Y-axis: Ownership Type. Add Facility Size as Small Multiple (Format, Small Multiples). This produces three side-by-side charts one per size band, directly testing whether the For-Profit gap exists uniformly or only in large facilities.
2	**100% Stacked Bar for Composition** 100% Stacked Bar. X-axis: Ownership Type. Values: Average RN Hours and Average Total Hours. A smaller RN slice in For-Profit bars confirms the cost-substitution hypothesis visually.
3	**Constant Line at 0.75** Analytics pane, Constant Line at 0.75, labelled Minimum RN Guideline. Makes it immediately visible which ownership types fall below the recommended floor.
4	**Conditional Table** Table with Ownership Type, Facility Size, Average RN Hours, RN Share of Total Hours, and RN Hours Gap. Apply background conditional formatting to the Gap column: red for negative, green for positive.

BP3 Staffing Rating as a Predictor of Overall CMS Rating

Business Question: How reliably does CMS Staffing Rating predict Overall Rating and which ownership types show the strongest versus weakest alignment?

Key Columns: Staffing Rating, Overall Rating, Ownership Type

Hypothesis: The correlation will be positive but imperfect, with meaningful outliers where facilities score high on staffing but low overall due to health inspection or QM failures.

DAX Measures

Rating Gap	`Rating Gap = AVERAGE([Overall Rating]) - AVERAGE([Staffing Rating])`
Outlier Flag	`Outlier Classification = IF(ABS([Rating Gap]) > 1, "Rating Divergence", "Aligned")`

Build Steps

1	**Scatter Plot with Per-Series Trend Lines** Scatter Chart. X: Staffing Rating. Y: Overall Rating. Values: Facility Name (each dot is a facility). Legend: Ownership Type. Analytics pane: add a Trend Line per series. A steeper line for Non-Profit indicates a stronger staffing-to-overall relationship in that ownership type.
2	**Quadrant Labels via Text Boxes** Overlay four Text Box labels on the scatter quadrants: High Staffing Low Overall (inspection failures), Low Staffing High Overall (QM compensating), High on Both, and Low on Both. This makes the scatter analytically legible for a non-technical audience.

3	**Drill-Through Page: Outlier Facility Detail** New page called Outlier Detail. Table with Facility Name, State, Ownership Type, Staffing Rating, Overall Rating, and Rating Gap measure. From the scatter users can right-click any dot to see that facility's full data row.

4	**Supporting Matrix Cross-Tab** Matrix with Staffing Rating as Rows and Overall Rating as Columns. Values: Count of Facility Name. Apply background colour formatting. The diagonal where both ratings are equal should be densest. Cells far from the diagonal are outliers.

BP4 Weekend Staffing Gap: Quantifying the Care Continuity Risk

Business Question: How large is the reduction in nurse staffing over weekends by state and ownership type and which combination produces the highest risk?

Key Columns: State, Ownership Type, Total Nurse Hours Per Resident Day, Weekend Nurse Hours Per Resident Day

Hypothesis: For-Profit facilities in high-volume states will show the largest weekend gaps as cost-minimizing incentives are strongest during lower-oversight weekend periods.

DAX Measures

Weekend Gap	`Weekend Gap Hrs = AVERAGE([Total Nurse Hours Per Resident Day]) - AVERAGE([Weekend Nurse Hours Per Resident Day])`

Gap Severity	`Gap Severity = SWITCH(TRUE(),[Weekend Gap Hrs]>0.75,"Critical",[Weekend Gap Hrs]>0.5,"High",[Weekend Gap Hrs]>0.25,"Moderate","Low")`

Build Steps

1	**Decomposition Tree** Add a Decomposition Tree visual (AI Visuals category). Set Analyse to Weekend Gap Hrs. Set Explain By to State, then Ownership Type. This allows interactive drilling from state to ownership type to see where the gap is driven. It is the most sophisticated visual in the report.

2	**Heat Map Matrix** Matrix. Rows: State (top 20 by gap). Columns: Ownership Type. Values: Weekend Gap Hrs. Apply white-to-red conditional formatting. High-gap cells turn deep red, making risk hotspots immediately identifiable.

3	**Gap Severity Slicer** Slicer based on Gap Severity measure. Set to Tile display mode. Clicking Critical filters all other visuals to show only the highest-risk combinations.

4	**Smart Narrative Insert** Insert a Smart Narrative visual below the charts. Power BI auto-generates a sentence summarising the current filter context. This demonstrates awareness of Power BI's AI-assisted narrative capabilities.

BP5 Turnover Risk Matrix: Ownership and Size as Compounding Retention Drivers

Business Question: Is high nursing staff turnover concentrated in specific ownership-size combinations and does the RN turnover gap signal a leadership-level retention failure?

Key Columns: Ownership Type, Facility Size, Total Nursing Turnover Rate, RN Turnover Rate

Hypothesis: Large For-Profit facilities will show the highest total turnover and the largest gap between total and RN turnover, indicating that RN retention failures are disproportionately severe at scale.

DAX Measures

Turnover Risk Index	`Turnover Risk Index = ([Total Nursing Turnover Rate] * 0.5) + ([RN Turnover Rate] * 0.5)`

RN Retention Pressure	`RN Retention Pressure = [RN Turnover Rate] - [Total Nursing Turnover Rate]`

Build Steps

1	**Conditional Colour Matrix** Matrix. Rows: Ownership Type. Columns: Facility Size. Values: Average Total Turnover, Average RN Turnover, and Turnover Risk Index. Apply background conditional formatting on all three fields using a green-yellow-red gradient. A 3x3 matrix with three value columns is a dense, high-information visual that signals analytical seniority.
2	**Waterfall Chart** Waterfall Chart. Category: Ownership Type, then Facility Size. Values: Turnover Risk Index. Waterfall charts show cumulative contribution, making it clear which segments add the most risk to the overall picture. This is an underused chart type that signals portfolio-grade knowledge.
3	**RN vs Total Turnover Scatter** Scatter Chart. X: Total Nursing Turnover Rate. Y: RN Turnover Rate. Legend: Ownership Type. Add a 45-degree Constant Line. Facilities above the line have disproportionately high RN turnover relative to total, signalling an RN-specific crisis separate from general workforce instability.
4	**Custom Tooltip Page** Create a Tooltip page (Format, Page Information, set as Tooltip). Add a small Card showing Facility Name, State, and Turnover Risk Index. Hover over any dot in the scatter chart to see this custom tooltip. This demonstrates a feature most analysts never configure.

PART 4 Microsoft Visio Deliverables

Execution Phase | Four Diagrams Demonstrating PM and Business Process Fluency

The four Visio diagrams together constitute the process and project management layer of the portfolio. They demonstrate that you understand not only data analysis but the structured project context in which that analysis is delivered. Build them in the order listed below, as each informs the next.

4.1 Project Lifecycle Swimlane (5 PMP Phases)

This is the highest-impact Visio deliverable for demonstrating PM knowledge. It shows all five PMBOK phases as horizontal role swimlanes and maps specific project deliverables to the responsible role within each phase. Decision diamonds at approval gates (Charter Approved, Scope Signed Off, Visuals Validated) show that you understand governance checkpoints, not just task sequencing.

Visio Setup: File, New, Flowchart, Cross-Functional Flowchart. Add horizontal swimlanes for: Project Manager, Data Analyst, Stakeholder Reviewer, Power BI Layer, and Visio Layer. Place deliverable shapes in the correct lane and phase column. Add Diamond decision shapes at key approval gates. Connect all shapes with arrows. Cross-lane arrows represent handoffs between roles. Apply a clean theme from the Design tab.

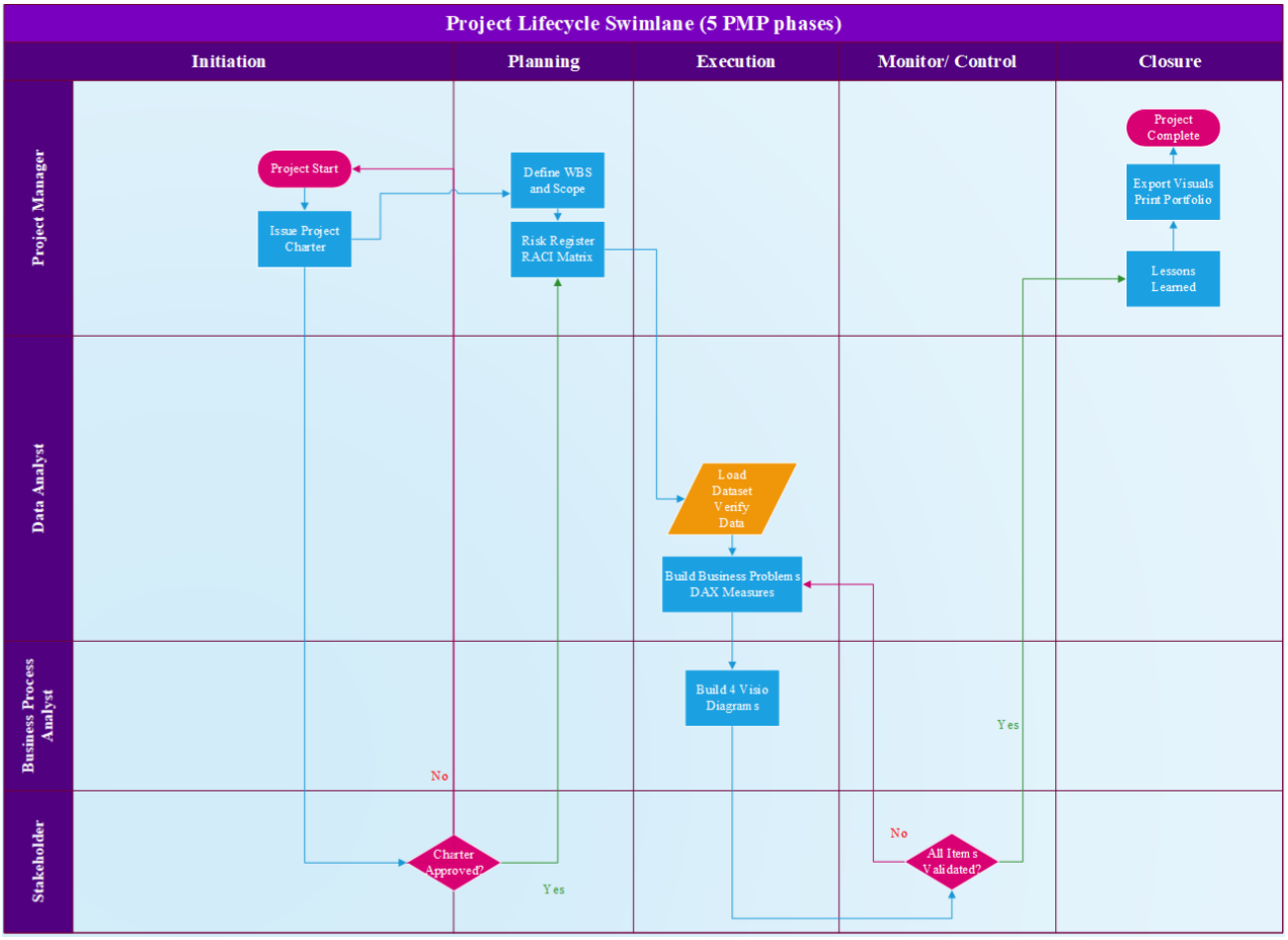

Figure 2: Project Lifecycle Swimlane – 5 PMP Phases with Role Swimlanes (Reference: Build in Visio Cross-Functional Flowchart template)

4.2 WBS Hierarchical Tree Diagram

The WBS tree is the visual format recommended in PMBOK for presenting work breakdown structure to stakeholders. It is significantly more readable than a table in a presentation context because the hierarchy is immediately visible without reading row by row. Each shape contains its WBS code and deliverable name. Colours are applied consistently by phase. Diagram Presented in Figure 1 (Project Management Plan section).

Visio Build Instructions: File, New, Business, Org Chart Wizard or Basic Flowchart. Place the root node at the top: 0.0 NH Staffing Analytics Portfolio Project. Add five Level 1 child shapes below for each PMP phase (1.0 Initiation through 5.0 Closure). Add Level 2 grandchild shapes for each WBS work package. Include WBS codes inside each shape. Apply Auto Layout under Home, Position, Auto Layout to balance the tree. Apply consistent colours by phase using Format Painter.

4.3 Gantt Chart (21-Day Sprint, March 1–21 2026)

The Gantt chart is the most universally recognized project management scheduling artefact. It shows tasks on a timeline, making schedule dependencies and phase boundaries visible at a glance. Five milestone diamonds are placed at each M1 through M5 completion date. The colour coding per phase matches all other project documents.

Visio Setup: File, New, Schedule, Gantt Chart. Set Start Date to 1 March 2026, Finish Date to 21 March 2026, time unit to Days. Rename task rows using WBS task names. Drag bar edges to set duration. Add five milestone diamond shapes at the correct dates. Apply phase colour coding. Note: Monitoring and Control runs in parallel with Execution, not after it. This is a common misconception and making this overlap visible on the Gantt signals genuine PMP understanding.

ID	Task Name	Start	Finish	Duration
1	Project Charter	2026-03-02	2026-03-03	2d
2	Stakeholder Register	2026-03-03	2026-03-04	2d
3	Scope Statement	2026-03-04	2026-03-05	2d
4	WBS + Dictionary	2026-03-05	2026-03-06	2d
5	Risk Register	2026-03-06	2026-03-09	2d
6	Communication Plan	2026-03-06	2026-03-09	2d
7	RACI Matrix	2026-03-09	2026-03-10	2d
8	Milestone Schedule	2026-03-09	2026-03-10	2d
9	Load Dataset / Power BI	2026-03-11	2026-03-11	1d
10	Build BP1 + BP2 Visuals	2026-03-11	2026-03-12	2d
11	Build BP3 + BP4 Visuals	2026-03-12	2026-03-13	2d
12	Build BP5 + DAX Measures	2026-03-13	2026-03-16	2d
13	Conditional Formatting	2026-03-16	2026-03-16	1d
14	Visio Lifecycle Swimlane	2026-03-17	2026-03-17	1d
15	Visio WBS + Gantt	2026-03-18	2026-03-18	1d
16	Visio Workflow Flowchart	2026-03-18	2026-03-18	1d
17	Validate All Visuals	2026-03-18	2026-03-19	2d
18	Change Control Review	2026-03-19	2026-03-19	1d
19	Export + Print Portfolio	2026-03-20	2026-03-20	1d
20	Lessons Learned	2026-03-20	2026-03-20	1d

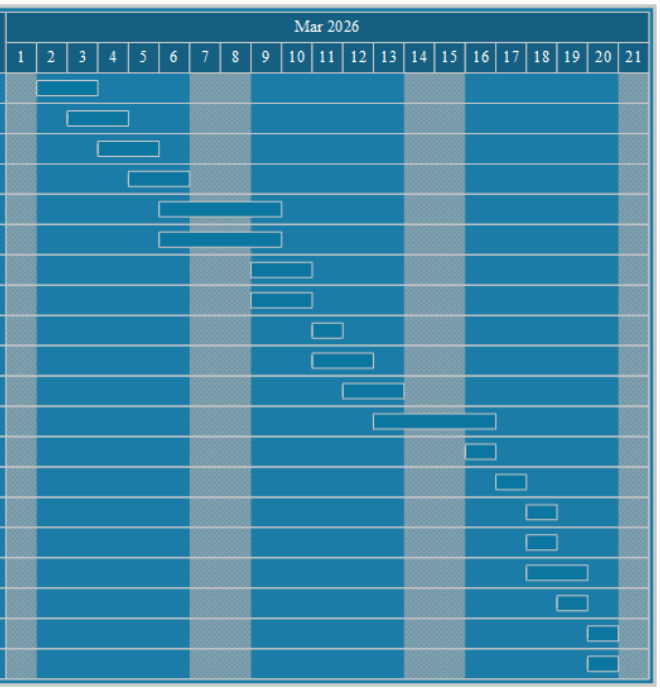

Figure 3: Project Gantt Chart – 21-Day Sprint with 5 Phase Milestones, 1 Mar to 21 Mar 2026 (Build in Visio Schedule > Gantt Chart template)

4.4 Analytics Workflow Flowchart

This flowchart zooms into the Execution phase and maps the step-by-step analytical workflow from receiving the dataset to printing the final portfolio. It is the deliverable most directly aligned to the Business Process half of the role title. Decision diamonds at key validation gates show that the workflow includes quality checkpoints, not just sequential tasks.

Visio Setup: File, New, Flowchart, Basic Flowchart. Start with an Oval labelled START: Receive Dataset. Progress through Rectangles for each major step. Add Diamond decision shapes at: Does the file load without errors? Does BP1 display correctly? Are all 5 pages validated? Connect each No branch to a loop-back with a label such as Fix and Retry. End with an Oval labelled END: Portfolio Ready. Apply a clean blue theme.

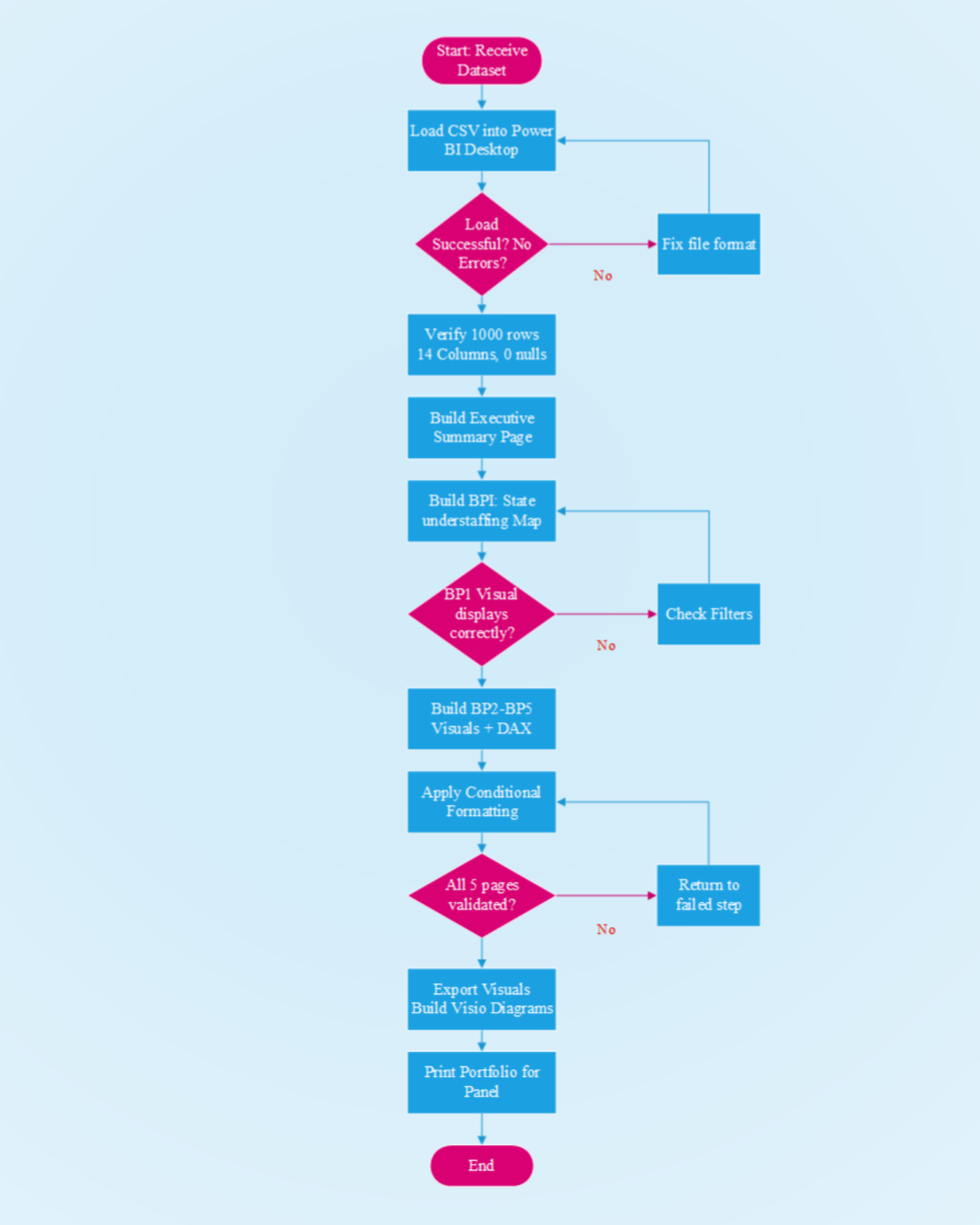

Figure 4: Analytics Workflow Flowchart – Execution Phase Zoom-In (Build in Visio Basic Flowchart template)

PART 5 Project Closure

Monitoring, Control, and Closing Phase | Validation, Lessons Learned, Portfolio Assembly

5.1 Validation Checklist

In PMBOK this activity is called Quality Control. A deliverable is only accepted when it passes its defined acceptance criteria. Mark each item as Pass or Fail before assembling the print portfolio.

Deliverable	Acceptance Criterion	Status
Project Charter	All eight fields populated. Objective is clear and measurable.	[ ] Pass
Scope Statement	In Scope and Out of Scope explicitly listed. Creep table has five scenarios.	[ ] Pass
WBS Tree Diagram	All WBS codes present. Hierarchy shows three levels. Colours match phases.	[ ] Pass
Risk Register	Seven risks logged with Probability, Impact, Score, and Response.	[ ] Pass
Assumption Log	Six assumptions listed. Each has a consequence if false.	[ ] Pass
Stakeholder Register	Five stakeholders identified with engagement strategy per role.	[ ] Pass
RACI Matrix	All deliverables have at least one R and one A assigned.	[ ] Pass
Communication Plan	Six communication events defined with channel and role.	[ ] Pass
Change Control Log	Template complete. At least one example entry present.	[ ] Pass
Power BI Executive Summary	Four KPI cards, two synced slicers visible across all pages.	[ ] Pass
Power BI BP1	Choropleth map, dual-axis bar, and drill-through page all present.	[ ] Pass
Power BI BP2	Small Multiple bar chart, 100% stacked bar, and reference line present.	[ ] Pass
Power BI BP3	Scatter with per-series trend lines, outlier labels, and matrix present.	[ ] Pass
Power BI BP4	Decomposition Tree, heat map matrix, and severity slicer present.	[ ] Pass
Power BI BP5	Conditional matrix, waterfall chart, scatter with custom tooltip page present.	[ ] Pass
Visio Swimlane	Five phase columns, five role lanes, decision gates visible.	[ ] Pass
Visio WBS Tree	Root node, five L1 phase nodes, Level 2 work packages, WBS codes inside shapes.	[ ] Pass

Visio Gantt	21-day timeline 1–21 Mar, five milestones, phase colour coding, WBS task names.	[ ] Pass
Visio Workflow Flowchart	Start and End ovals, decision diamonds at all validation gates, loop-back arrows.	[ ] Pass
Print Portfolio	Three copies, colour printed, ordered per print guide, assembled in folder.	[ ] Pass

5.2 Lessons Learned Register

Capturing Lessons Learned throughout the project, not only at closure, produces more accurate and actionable insights. In PMBOK 7th Edition the Lessons Learned Register is an ongoing artefact that is transferred to the Organizational Process Assets repository at project close.

Category	Lesson Observed	Recommendation for Future Projects
Scope Control	Defining the Out of Scope section before execution began prevented the temptation to add a sixth business problem mid-sprint	Always write an explicit Out of Scope section in the Scope Statement before any execution activity begins
Execution: Power BI	The Decomposition Tree visual took longer to configure than estimated because of AI visual licensing requirements	Add 30 minutes buffer to any AI visual or experimental chart type when scheduling Power BI build tasks
Execution: Visio	Auto Layout in Visio significantly reduced formatting time for the WBS tree and should be applied before any manual adjustments	Always apply Auto Layout before making visual adjustments in Visio. Manual adjustments on an unbalanced tree waste significant time.
Risk Management	The printer contingency risk seemed unlikely but was worth planning for, as print quality issues on interview day would be high impact	Low-probability high-impact risks deserve response plans regardless of how unlikely they appear at planning time
Stakeholder Management	Framing every design decision around what the panel would see and need made the portfolio more purposeful	Always name the audience and their evaluation criteria before designing any deliverable, not after
Communication	Rehearsing the verbal walkthrough on Day 20 identified two gaps in the analytical narrative that were corrected before interview day	Schedule at least one full verbal rehearsal against printed copies before any stakeholder presentation
Technical: DAX	All DAX measures should be validated in a Table visual before embedding in a chart to prevent filter context errors	Test every DAX measure in isolation before adding it to a chart. Filter context errors are invisible until the wrong context is applied.

End of Document | Version 1.0 | Project Timeline: 1 March to 21 March 2026